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Techno-economical study of hybrid power system for a remote village in Algeria

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Abstract

The share of renewable energy sources in Algeria primary energy supply is relatively low compared with European countries, though the trend of development is positive. One of the main strategic priorities of NEAL (New Energy Algeria), which is Algeria's renewable energy agency (government, Sonelgaz and Sonatrach), is striving to achieve a share of 10–12% renewable energy sources in primary energy supply by 2010.This article presents techno-economic assessment for off-grid hybrid generation systems of a site in south western Algeria. The HOMER model is used to evaluate the energy production, life-cycle costs and greenhouse gas emissions reduction for this study. In the present scenario, for wind speed less than 5.0 m/s the existing diesel power plant is the only feasible solution over the range of fuel prices used in the simulation. The wind diesel hybrid system becomes feasible at a wind speed of 5.48 m/s or more and a fuel price of 0.162$/L or more. If the carbon tax is taken into consideration and subsidy is abolished, then it is expected that the hybrid system will become feasible. The maximum annual capacity shortage did not have any effect on the cost of energy, which may be accounted for by larger sizes of wind machines and diesel generators.

Keywords: HOMER; Hybrid power system; Remote village; Techno-economical study

1. Introduction

Most scientists now agree that the human-induced global climate change poses a serious threat to both society and the Earth's ecosystems. Renewable energy holds the key to future prosperity and a healthy global environment and is considered as a promising way to solve the problems of environmental pollution, such as major environmental disasters, water pollution, maritime pollution, land use and sitting impact, radiation and radioactivity, solid waste disposal, hazardous air pollutants, ambient air quality (CO, CO_2, SO_x, NO_x effluent gas emissions), acid rain, stratospheric ozone depletion, and global warming. Much of this capacity will be installed in developing nations, where solar and wind electric power is already competitive.

Currently, wind energy is generally developed in the industrialized countries for a number of reasons including growing environmental concerns and relatively high price of electricity generated. The total world wind power capacity reached up to 72,628 MW in 2006. Wind power is present, today, in the energy mix of more than 60 countries, not only in practically all of the developed countries, but also in more and more of the developing ones. India already has a 6053 MW installed capacity, which represents the fourth place position behind Germany, Spain and the USA. China enters the top 10 (in the eighth place) with 1699 MW installed capacity.

The commercial wind turbine market evolved from domestic and agricultural applications of small machines in the few kilowatt size range to utility interconnected wind farm applications of intermediate-scale machines of

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megawatts. The power supply for the remote and isolated areas mainly depends on "stand alone" diesel or hybrid generation systems, which are now a possible economical alternative to running the grid all the way to these areas. Many rural communities, who are in need of drinking water and water for livestock or irrigation, do not have access to grid electricity.

So wind–diesel hybrid system is the favorable and sensible choice for remote village electrification in many localities such as Denmark, India, Morocco, Libya, Egypt, Sultanate Oman, Kingdom of Bahrain, etc. Remote applications may include an electricity supply to military installations, desalination plants, gas stations, pipeline cathodic protection, telecommunication stations, alpine huts, or remote stations for data logging of environmental parameters, irrigation and water pumping, and agribusiness operations, etc. [1].

Many literatures reported to determine the optimum hybrid energy system for small loads (ranging from few watts to few kilowatts) in a given location [2–6]. The reported studies show that the renewable energy-based off-grid hybrid generation systems can compete with power from the gird in remote locations, where the grid is either not feasible or nonexistent.

Moreover, the hybrid systems such as wind/diesel are now proven technologies and are an option to supply small electrical loads at remote locations as reported by Zhang Hongyi et al. [8], Lundsager and Bindner [7]. In developing countries, interest in medium- to large-scale wind–diesel hybrid power system for rural electrification has grown enormously among energy officials and utility planners [24]. Bonanno et al. [9] presented a logistical model for hybrid generation system relative to a small isolated community, an island in the Mediterranean Sea. This study revealed that we can evaluate the prediction of energetic performance, in order to select good management strategies, as well as for an optimal sizing of the generating groups and the energy storage.

With growing global awareness of the need for clean sources of energy, wind energy in particular, much research on a lot of scientific study is being carried out in Saudi Arabia [14–22]. Rehman et al. conducted a study to perform an economical feasibility assessment of an existing grid-connected diesel power plant supplying energy to a remotely located village by adding wind turbine/s in the existing power system in order to reduce the diesel consumption and environmental pollution, using the HOMER simulation model. They found that the wind diesel hybrid system becomes feasible at a wind speed of 6.0 m/s or more and a fuel price of 0.1$/L or more [24].

In Algeria, work on wind resource assessment dates back to 1990 when a wind atlas was published by Hammouche [25], giving the results of the statistical study of 37 locations using the WASP. Kasbadji Merzouk collected mean wind data from 64 stations for wind energy potential of Algeria at a height of 10 m. It is found that the windy regions are located at the south west of Algeria, Sahara [26].

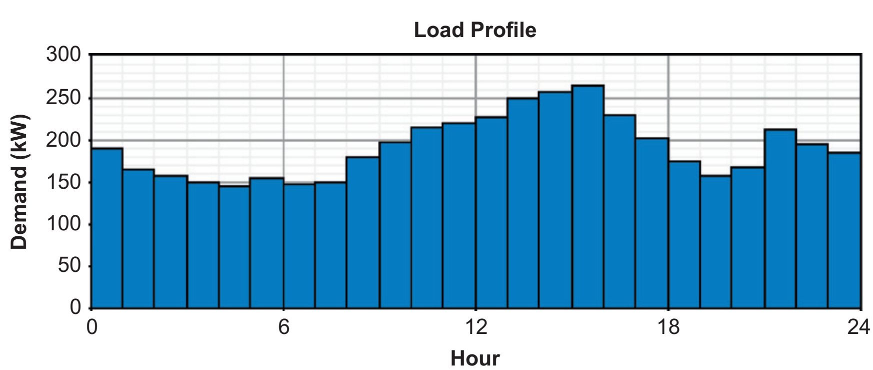

Fig. 1. Hourly load variation at Oum El Assal for the day August 5, 2006.

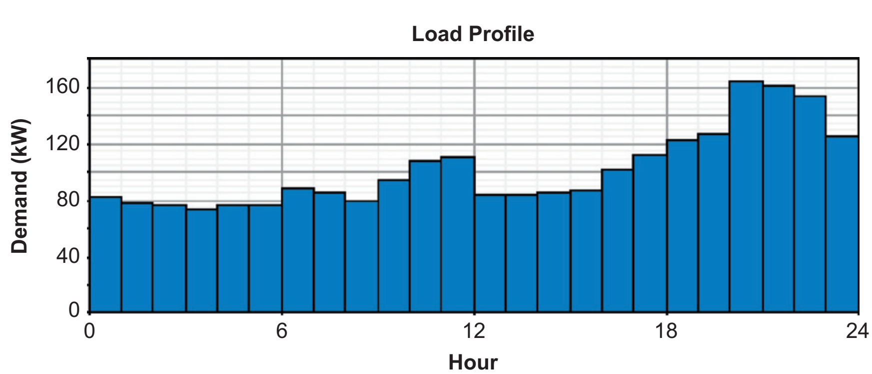

Fig. 2. Hourly load variation at Oum El Assal for the day April 17, 2006.

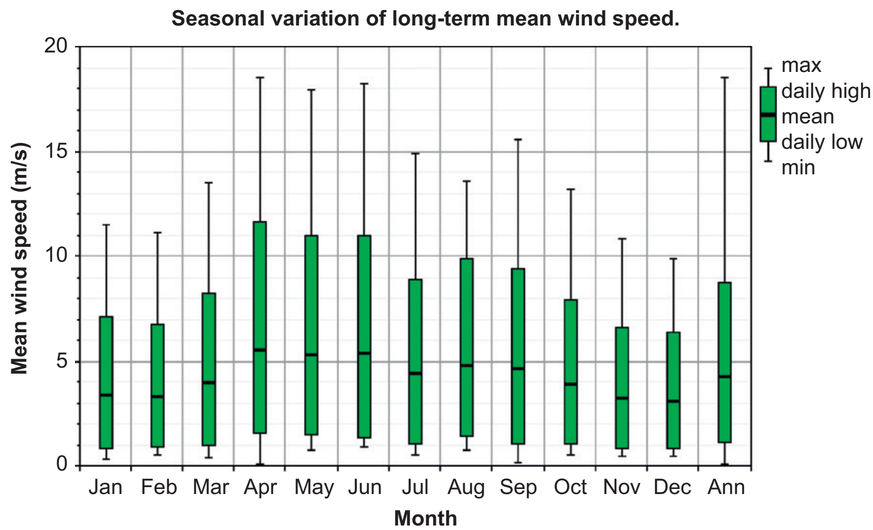

Fig. 3. Seasonal variation of long-term mean wind speed.

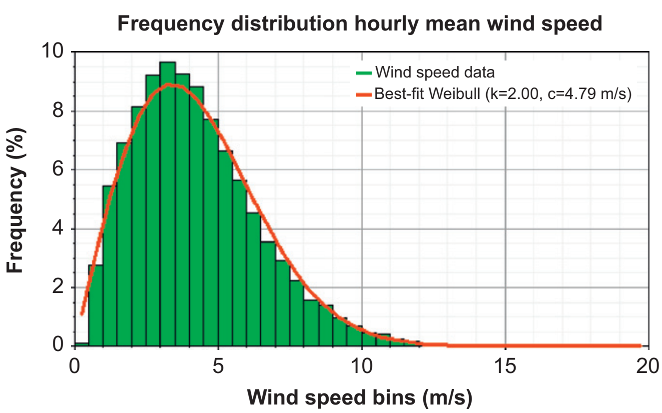

Fig. 4. Frequency distribution hourly mean wind speed.

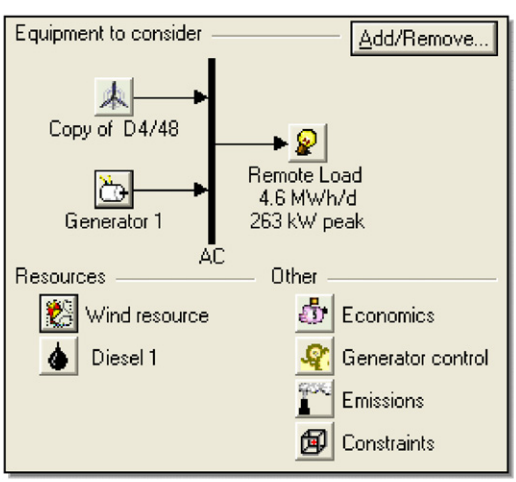

Fig. 5. Wind–diesel energy system for an isolated grid.

The aim of the present study is to carry out an economical feasibility of an existing grid-connected diesel power plant supplying energy to a remotely located village of about 425 consumers by adding wind turbine in the existing power system to reduce the diesel consumption and environmental pollution.

Table 1
Diesel generator information from diesel power plant

Parameters	Value/information
Manufacturer	Cummins
Rated power	500 kW
Minimum allowed power	74 kW (according to the minimum load curve)
No load fuel consumption	36 L/H (assumed)
Full load fuel consumption	140 L/H (assumed)
Power factor	0.8
Voltage	400 V
Rated current	903 A
Frequency	50 Hz
Rotating speed	1500 rpm
Battery voltage	24 V

2. Electric load data of the village

The energy demand of the village is shown in Fig. 1. The annual peak load of 263 kW was observed on August 05 at 15:30 h and a minimum of 74 kW on April 17 at 03:30 h as shown in Fig. 2. So the higher demand exists between April and September, while relatively smaller load requirements are found during rest of the period of the year.

3. Wind speed data

The wind speed data were taken over a period of 8 years at a height of 10 m above ground level (AGL).

Since the rotors of the modern wind turbines are placed at heights varying between 40 and 110 m, these data were calculated at 60 m hub height using 1/7 power law. At this height the average wind speed became 5.48 m/s, while at 10 m it was only 4.25 m/s. As seen from Fig. 3,

Table 2
Diesel generator cost data obtained locally

Manufacturer	Rating (kW)	Capital cost ($)	Replacement cost ($)	*O* and *M* cost ($/h)
Cummins	500	80,355	53,570	3.01

Table 3
Fuel cost and technical data

Parameters	Value
Cost	0.162$/L (11.94 DA/L include transportation)
Lower heating value	45.62 MJ/kg (10.9 therm/kg)
Density	831 kg/m^3 (0.831 kg/L)
Carbon content	80%

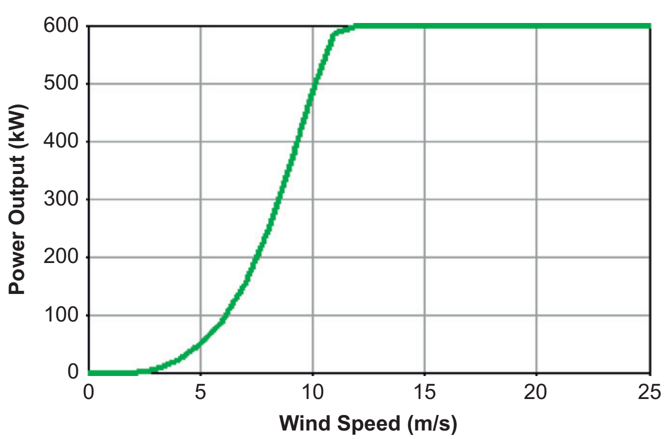

Fig. 6. Wind power curve of D4/48 wind turbine from DeWind of 600 kW rated power.

Table 4
Wind turbine technical data

Parameters	Value/information
Manufacturer	DeWind
Hub height	60 M
Rotor diameter	48 M
Cut-in-speed	3.0 m/s
Cut-out-speed	22.0 m/s
Rated speed	11.5 m/s
Rated power	600 kW
Overhaul period	25 years
Wind power scale factor	1
Wind power response factor	1.5
Voltage	690 V
Wind turbine capital cost ($)	575,000 (assumed)
Replacement or overhaul cost	400,000 (assumed)

the long-term seasonal wind speeds were found to be relatively higher during March–September compared to other months. In general, higher winds were observed between 09:00 and 18:00 h, and lower during the rest of the period of the day. This indicates that higher electricity could be produced during 09:00–18:00 h, which also coincide with higher electricity demand time.

The frequency distribution of wind data shows that wind remained at 3.5 m/s and below it for about 31% of the time during the entire year and above it for the rest of the period, as shown in Fig. 4. This means that the energy could be harnessed for almost 69% of the time using wind machines with a cut-in-speed of 3.5 m/s or more.

Table 5
Economic inputs

Parameters	Value (%)
Percent of annual peak load	0
Percent of hourly load	10
Percent of hourly solar output	0
Percent of hourly wind output	40

Table 6
Constraints inputs

Parameters	Value (%)
Maximum unserved energy	0
Minimum renewable fraction	0, 5, 10, 15, 20 and 25
Minimum battery life	N/A
Maximum annual capacity shortage	0, 3, 5, 7 and 10

4. System components

The main components of an isolated grid-connected wind–diesel hybrid system are diesel generators and wind turbines. A typical wind–diesel hybrid system used in HOMER software is shown in Fig. 5. The system consists of one diesel generator, the wind turbines, an AC bus and the primary load. The cost of each component, the number of units used in simulation, the economical and control parameters required by the software are discussed in the forthcoming paragraphs.

4.1. Diesel generator

The diesel power plant at the remote village consists of four diesel units of 500 kW each. At present, of these four units one is sufficient to meet the load of the village. In fact, it is a new power plant that was installed in 2006, to satisfy the need of the electric load of the village. The technical information on diesel-generating units, being used in the village, was obtained from the power plant. The details of various parameters are given in Table 1. The cost data, obtained from SPE (Société de Production d'Electricité) subsidiary company of SONELGAZ communication, are given in Table 2.

The operation and maintenance cost of 3.01$/h was used in the simulation run. The fuel cost obtained locally, including the transportation cost, was 0.102$/L, as given in

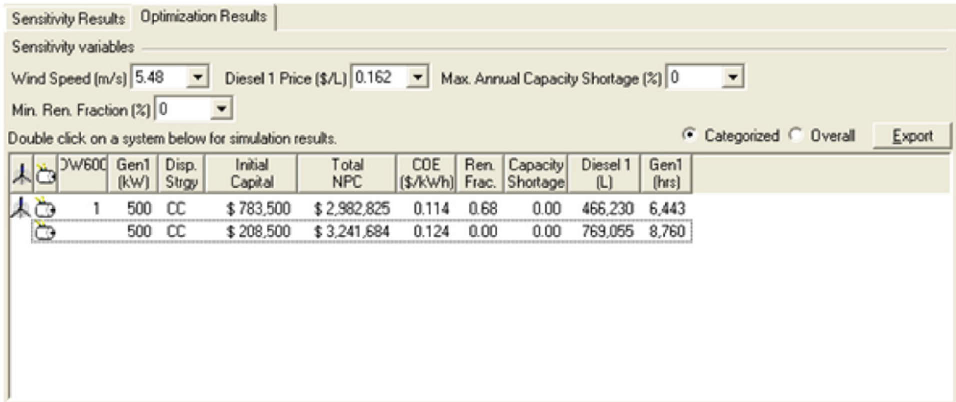

Fig. 7. Optimization results for wind speed of 5.48 m/s, diesel price of 0.162$/L maximal annual capacity shortage 0% and minimum renewable fraction 0%.

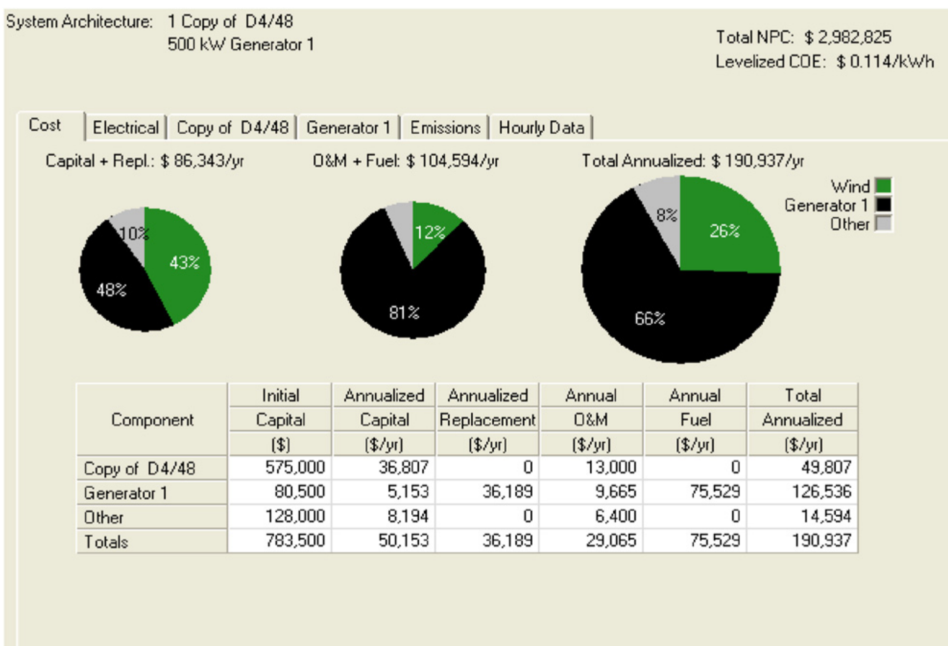

Fig. 8. Wind–diesel hybrid power system cost analysis for maximal annual capacity shortage 0% and minimum renewable fraction 0%.

Table 3. This table also includes the other technical information related to diesel fuel.

4.2. Wind turbine

The other major component of the wind-diesel hybrid system after the diesel-generating set is the wind energy conversion system or wind turbine. The modern wind machines are very efficient and are found in big sizes, capacity wise. Today's market standard size of the wind machine is greater than 1.5 MW. The rotor diameter of these machines varies between 40 and 120 m and tower height between 40 and 110 m and more.

The modern wind turbine produces more energy due to high wind speeds at higher hub heights, since the energy yield from the wind energy conversion system depends on the availability of wind and its variation. In the present case, WECS of 600 kW from DeWind is used. The wind power curve of the D4/48 wind machine, obtained from [32], is shown in Fig. 6. The technical and cost information of the wind machine is summarized in Table 4. The operation and maintenance cost of $13,000 per wind turbine per year has been assumed based on literature data.

4.3. Economic and constraints inputs of the system

The lifetime project was taken as 25 years and the annual real interest rate as 4%. The fixed capital cost of the system was assumed to be US$128,000, while fixed *O* and *M* costs as US$6400 per annum. Furthermore, the capacity shortage penalty was not considered. The economic and constraints inputs are given in Tables 5 and 6, respectively.

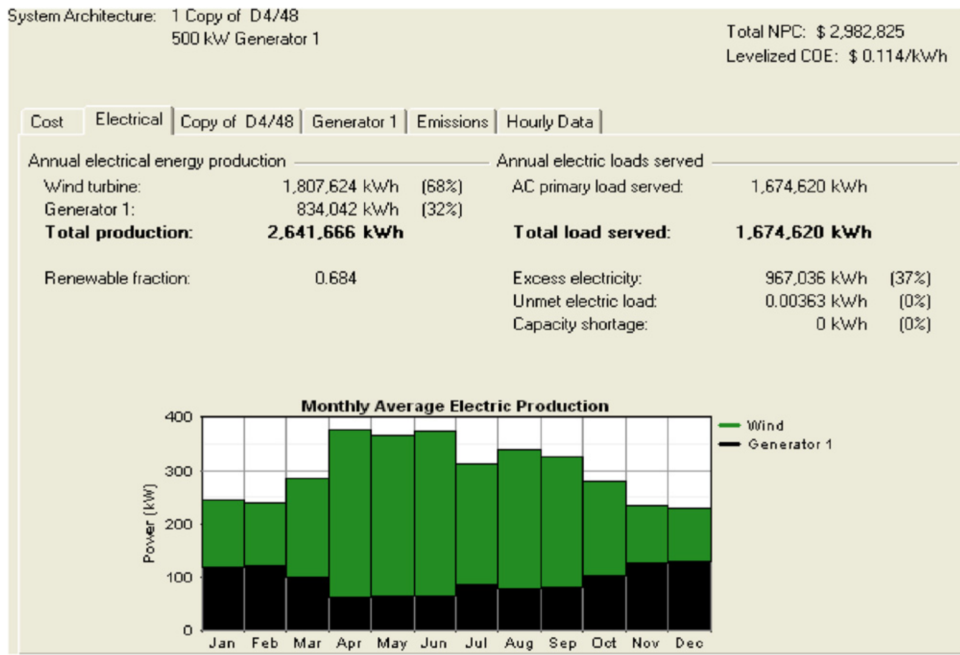

Fig. 9. Energy yield from wind machine and diesel generator.

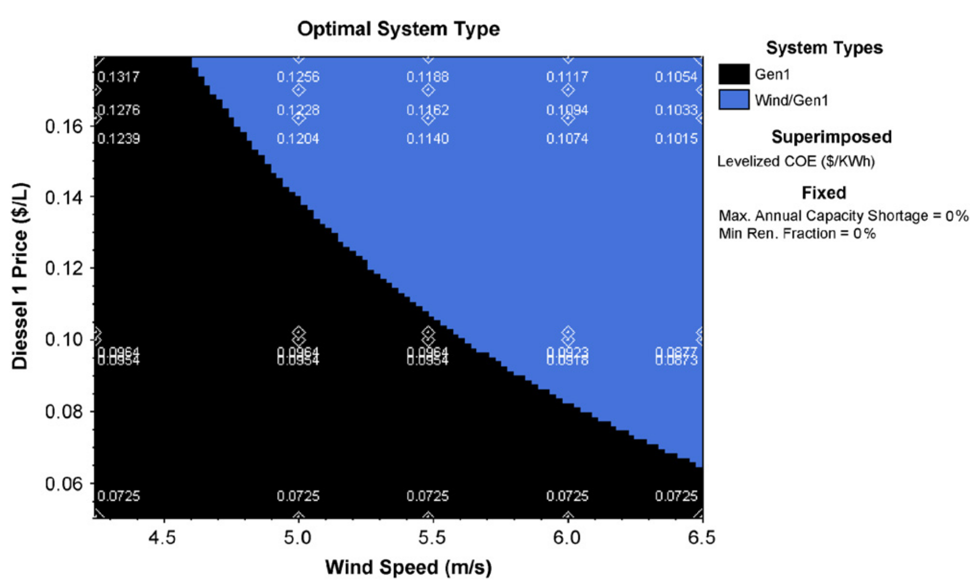

Fig. 10. Optimized wind–diesel hybrid system for MACS = 0% and MRF = 0%.

5. Results and discussion

HOMER provides the results in terms of optimal systems and the sensitivity analysis. In this software the optimized results are presented categorically for a particular set of sensitivity parameters like wind speed, maximum annual capacity shortage (MACS), minimum renewable fraction (MRF) and fuel price in the present case. The simulation results are presented in the following paragraphs.

5.1. Optimization results

The optimization results for a wind speed of 5.48 m/s, MACS of 0%, MRF of 0% and a fuel price of 0.162$/L are summarized in Fig. 7. In this case, a diesel power system seems to be most feasible economically with a minimum total net present cost (NPC) of 2,982,825$ and minimum cost of energy (COE) of 0.114$/kWh, although the system represents a higher initial capital. Generally at 5 m/s and more wind speed, the diesel

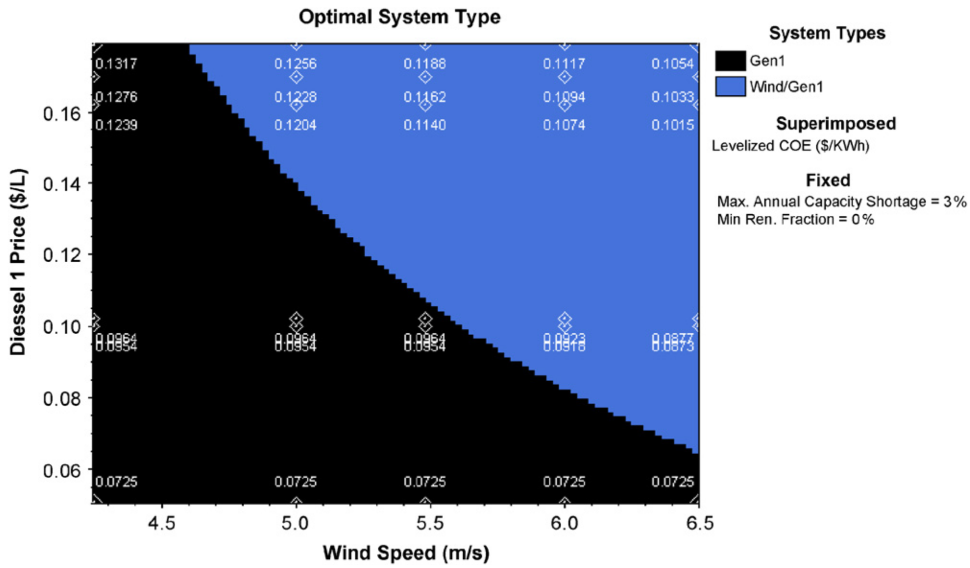

Fig. 11. Optimized wind–diesel hybrid system for MACS = 3% and MRF = 3%.

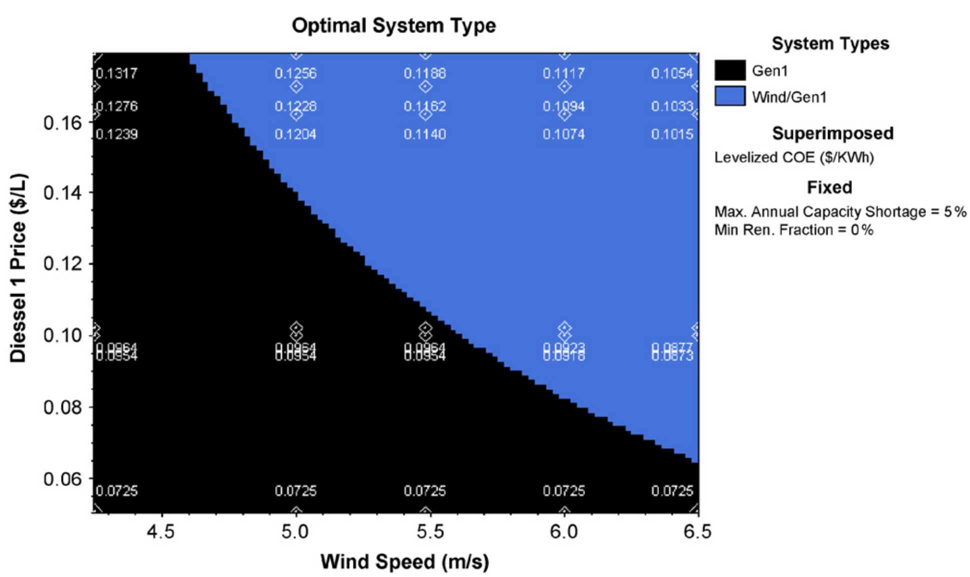

Fig. 12. Optimized wind-diesel hybrid system for MACS = 3% and MRF = 3%.

Pollutant	Emissions
Carbon dioxide:	1,865,169 kg/yr
Carbon monoxide:	4,999 kg/yr
Unburned hydrocarbons:	554 kg/yr
Particulate matter:	377 kg/yr
Sulfur dioxide:	4,122 kg/yr
Nitrogen oxides:	44,605 kg/yr

Fig. 13. GHG for diesel system Fig. 14. GHG for WDH system.

system was found to be most feasible solution with COE, less than 0.124$/kWh.

As seen from Fig. 8, almost about 60% of the initial cost of the wind–diesel hybrid system was accounted for diesel power system and even more than 90% of the *O* & *M* and fuel cost was accounted for diesel system. The total annualized cost for wind equipment accounted for 49,807$ (26% of the entire wind–diesel power plant cost), while for diesel power system 141,130$ (74% of the entire wind–diesel power plant cost). The energy yield from different components of the wind-diesel hybrid system is shown in Fig. 9 for the total primary energy requirement of the village; the wind machine produced 1,807,624 kWh

Pollutant	Emissions
Carbon dioxide:	1,130,736 kg/yr
Carbon monoxide:	3,030 kg/yr
Unburned hydrocarbons:	336 kg/yr
Particulate matter:	228 kg/yr
Sulfur dioxide:	2,499 kg/yr
Nitrogen oxides:	27,041 kg/yr

Fig. 14. GHG for WDH system.

(68% of the total energy served), while the diesel generators produced almost 32% of the energy i.e. 834,042 kWh. Although an excess energy of 967,036 kWh (37%) was produced, with no capacity shortage (0 kWh).

5.2. Sensitivity results

The HOMER eliminates all infeasible systems and presents the results in ascending order of NPC. In the present case, wind speed (4.24, 5, 5.48, 6 and 6.5), diesel price (0.05, 0.1, 0.102, 0.162, 0.170 and 0.179$/L), MACS (0, 3, 5, 7 and 10%) and minimum renewable energy fraction (0, 5, 10, 15, 20 and 25%) were used as sensitivity variables. The optimization results are shown in terms of wind speed and diesel price in Figs. 10–12 for 0, 3 and 10% MACSs, respectively. This type of graphical representation of optimal system-type provides information that a particular system will be optimal at certain wind speed and a certain fuel cost. Furthermore, the wind speed and diesel cost are usually site-dependent, so one can conclude that at a particular wind speed and fuel cost the system will be optimal for a particular place or location.

The system shown in Fig. 10 reflects that for 5.0 m/s wind speed or more and diesel price of 0.162$/L or more, the wind-diesel hybrid system becomes economically feasible. The COE of such a system was 0.120$/kWh. With a minimal compromise on MACS of 3% and 5% the wind-diesel hybrid system becomes feasible only above 5.48 m/s wind speed and a fuel cost of 0.162$/L or more, as can be seen from Fig. 11,12, respectively, the effect of MACS is not noticeable because of large load and larger sizes of the wind machines and diesel generators.

5.3. Green house gases (GHG) reduction

The GHG pollute the environment (air, water and soil), which ultimately adversely affects the life of human beings. An indirect or hidden cost, which is not taken into consideration while using fossil fuels, is paid by the human beings. The diesel power system being used at the village adds a total of 1,919,826 kg of pollutants into the local atmosphere of the village every year. The wind–diesel hybrid system can bring down the quantity of the pollutants to 1,163,870 kg per year. This shows a reduction of 755,956 kg (approximately 40%) of pollutants every year. The concentrations of various constituents of pollutants like CO_2, CO, nitrogen, etc. for diesel and hybrid system are summarized in tables, Figs. 13 and 14, respectively.

6. Conclusion

The aim of this study is to perform an economical feasibility of an existing grid-connected diesel power plant supplying energy to a remotely located village by adding wind turbine in the existing power system in order to reduce the diesel consumption and environmental pollution, using the HOMER simulation model. It was found that the wind–diesel hybrid system becomes feasible at a wind speed of 5.48 m/s or more and a fuel price of 0.162$/L or more. The MACS did not have any impact on the system optimization.

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